

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing**
- Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI Chatbot for Customer Support (2024)

Developed and implemented an AI-powered chatbot leveraging retrieval-augmented generation (RAG) to provide instant, accurate responses to customer queries based on a company knowledge base. Engineered and validated vector database structures using ChromaDB and OpenAI embeddings for high relevance retrieval. Continuously refined AI-generated prompts and responses to ensure technical accuracy and clarity, reducing human agent workload by 35% and significantly improving response time. Provided structured feedback on the AI-generated solutions and code to enhance chatbot performance and reliability in a remote, autonomous development environment.

Tech: Python, LangChain, OpenAI API, ChromaDB, RAG Architecture

Customer Sentiment Analysis Using NLP Techniques (2024)

Designed, tested, and validated a machine learning pipeline to classify customer sentiment from online reviews with 89% accuracy. Conducted comprehensive data preprocessing including tokenization, stopword removal, and TF-IDF feature extraction to ensure high quality inputs. Developed logistic regression classification models and performed thorough debugging and code validation to guarantee algorithmic correctness. Delivered detailed technical documentation and structured feedback on model performance metrics to facilitate continuous improvements aligned with project goals in a remote, part-time setting.

Tech: Python, NLTK, TF-IDF, Logistic Regression

Website Traffic Analytics and Conversion Rate Optimization (2024)

Analyzed complex web traffic data to identify bottlenecks and opportunities for optimization, employing SQL, Google Analytics, and Power BI for data extraction, segmentation, and visualization. Built and validated funnel analysis dashboards, tagging content by marketing funnel stages and traffic sources to inform decision-making. Provided structured technical feedback and tested SQL queries and dashboard functionalities to ensure data accuracy and clarity. Achieved a 17% increase in conversion rates and a 10% reduction in bounce rates, demonstrating strong software engineering and analytical rigor in a remote project environment.

Tech: SQL, Power BI, Google Analytics

Employee Attrition Predictive Modeling and HR Analytics Dashboard (2024)

Developed a predictive attrition model using logistic regression, validated algorithms for accuracy and robustness, and performed exploratory data analysis on employee demographics and work-related factors. Created comprehensive HR dashboards with Power BI and Python to monitor attrition trends, tagging key drivers such as salary dissatisfaction and overtime. Delivered precise technical evaluations and improvement recommendations, ensuring the clarity and correctness of analytical solutions. Results directly contributed to a 12% reduction in attrition rates through data-driven policy adjustments, all completed within a remote, self-directed project framework.

Tech: Python, SQL, Power BI, Logistic Regression

About :

Experienced Software Engineer specializing in AI-driven automation and large language models (LLMs). Skilled in reviewing and refining AI-generated prompts, responses, and code to ensure technical accuracy and quality. Proficient in validating algorithms, debugging, testing, and providing clear, structured feedback on software solutions. Strong background in developing AI applications using Python, JavaScript, and related technologies. Detail-oriented technical writer comfortable working independently in a remote, project-based environment. Committed to organizing and tagging content by topic, difficulty, and programming language to support benchmarking and assessment of AI model capabilities.

